

RMO(ORISSA) – 2001

Max Mark – 100

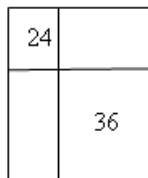
Time – 3 Hrs

Instructions:

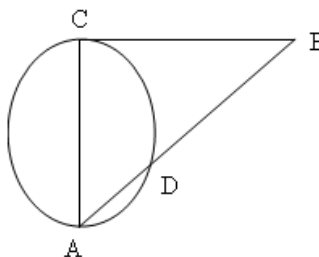
- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer as many questions as you can.
- Group A Q. 1 to Q.10 carry 3 marks each & Group B Q. 11 to Q. 17 carry 10 marks each.

Group – A

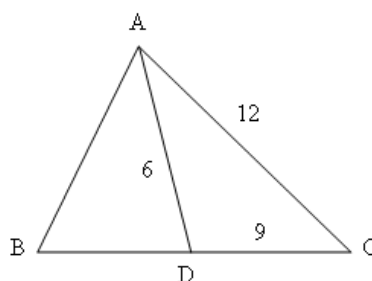
1. If 150% of x is y , then what % of $3x$ is $4y$?
2. For what smallest value of k , equation $\frac{3}{2} \cdot \frac{2}{1} \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \dots \cdot \frac{k}{k+1} \geq \frac{1}{8}$?
3. In the following figure a square is sub divided into 4 rectangles, out of which area of two rectangles are given as 24 and 36. If sides of rectangle are integers, then find the area of the square.



4. Find out an integer which when multiplied to 999999999 gives a number with all the digits 1.
5. Find the value of x , for $x[x] = 27$, where $[x]$ is the greatest integer less than or equal to x .
6. If $f(x) = 19x + 99$ and $f^{-1}(x) = ax + b$, then find value of $\frac{1}{a-b}$.
7. If number of subsets of a set A is 96 greater than the number of subsets of set B , then find the number of elements of set A .
8. In the figure AC is a diameter. BC is a tangent to the circle. If $AC = 5$, $BD = 2AD$, then find the length of BC .



9. In triangle ABC , $AC = 12$, $CD = 9$, $AD = 6$ and $\angle ABC = \angle DAC$. Find the perimeter of triangle ABC .



10. Find the ten's place digit of 2^{100} .

Group – B

11. The length of sides of a triangle DEF are inversely proportional to the altitudes of triangle ABC. Prove that lengths of altitude of triangle DEF are inversely proportional to length of sides of triangle ABC.
12. In triangle ABC, $a \geq b \geq c$ and area of triangle is 1 sq unit. Prove that $b \geq \sqrt{2}$.
13. I have purchased 72 marbles of same size to make a rectangular lawn of size 9ftX8ft. Cost price is less than Rs. 100 and it is same as $*0.6*$ where $*$ represents digits. Find the cost price of each marbles.
14. If a, b, c are sides of a triangle then prove that $\sqrt{a+b-c} + \sqrt{b+c-a} + \sqrt{c+a-b} \leq \sqrt{a} + \sqrt{b} + \sqrt{c}$.
15. If sum of first k natural number is a three digit number, then find the smallest value of k .
16. $*$ is a binary operation on A non empty set M , such that $(a*b)*b = a$ and $a*(a*b) = b$. Prove that $a*b = b*a$ for all $a, b \in M$.
17. Prove that $2^{3^n} + 1$ is not divisible by 17.

 **